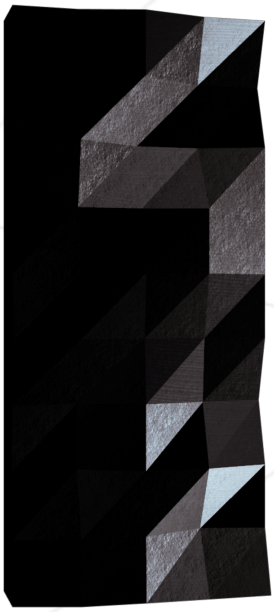

Ray Tracer

A graphical project

Summary: This project is nothing but a climax of amazing computer-generated images

Version: 4.1

Contents

I	**Foreword**	**2**
II	**AI Instructions**	**3**
III	**Objectives**	**5**
IV	**General Instructions**	**6**
V	**Mandatory part**	**7**
VI	**Options**	**8**
VII	**Bonuses**	**12**
VIII	**Submission and peer-evaluation**	**13**

Chapter I

Foreword

I have a little shadow that goes
in and out with me.
And what can be the use of him is
more than I can see.
He is very, very like me from the
heels up to the head;
And I see him jump before me,
When I jump into my bed.

The funniest thing about him is the way
he likes to grow
Not at all like proper children, which is
always very slow;
For he sometimes shoots up taller like an
India-rubber ball,
And he sometimes gets so little that
There's none of him at all.

One morning, very early, before the
sun I was up,
I rose and found the shining dew
on every buttercup;
But my lazy little shadow, like an
arrant sleepyhead.
Had stayed at home behind me and was
Fast asleep in bed.

My shadow
by Robert Louis Stevenson (1850-1894)

Chapter II

AI Instructions

Context

AI is now a powerful coding partner — alongside your peers — for tackling large and demanding projects. You will guide it through both technical and non-technical aspects of your work.

AI tools can boost your efficiency and improve the quality of your output, but you should be able to dive deep into any part of the project without relying on them.

Your AI partner supports you, but you remain fully responsible for making informed technical decisions and to clearly explain and defend them.

Main message

- Strive for a mature and responsible use of AI.
- Never let AI take responsibility for decisions — especially when it lacks awareness of your goals, constraints, or team dynamics.
- Maintain creativity, innovation, and human oversight through active collaboration with your peers. AI is trained on existing data and rarely generates truly new ideas.
- Stay informed about emerging trends and be ready to adapt to new concepts and technologies.

Learner rules:

- Maintain intellectual leadership over your projects and make your own informed decisions.
- Prioritise the collective intelligence of your team and peers.
- Actively stay informed about the ongoing evolution of AI technologies.

Phase outcomes:

- AI engineering skills.
- Increased efficiency.
- Greater reliability and quality.
- A pioneering mindset.

Comments and examples:

- Your peers can identify trade-offs, question assumptions, and help you improve. The first answer from an AI might not be the best — it may lack efficiency, security, or real added value. Now more than ever, you should rely on your peers.
- AI can make you faster, but your peers make you better. Collaboration, discussion, and mutual challenge are key to success.
- Be transparent about how AI was used in your projects, and clearly identify what was generated by AI tools.

✓ Good practice:

> I asked AI to help generate unit tests for my API. I reviewed them with my teammate, and we adjusted them for edge cases. It saved time, and we both learned something new.

✗ Bad practice:

> I had AI generate the entire architecture of my project. It "works," but when I'm asked to explain design decisions during the peer review or in front from of a customer, I cannot. I lose credibility and I fail.

Chapter III

Objectives

When it comes to render 3-Dimensions computer generated images, there are 2 possible approaches: “rasterization”, which is used by almost all graphic engines because of its efficiency on accelerated hardware, and the “ray tracing” method that is more extensive, more expensive, not adapted to real time production, but that comes with a high degree of visual realism.

Figure III.1: The pictures above are rendered with the ray tracing technique. Impressive isn’t it?

You may have mastered the basics of ray tracing with the `miniRT`. Now, the `RT` project opens the door of your second ray tracer coded in `C, C++ or rust`, awesome and functional.

And if you are all depressed reading this topic, think of the long way already paved by the lunatics from ILM, Pixar (and many more computer-generated images studio) that all have deployed infinite creativity to get better in that art. So, relax, take life easy and bound.

Chapter IV

General Instructions

- This project will be corrected by humans only. You are allowed to organise and name your files as you want.
- The executable file must be named `rt`.
- You're free to choose between C, C++ or Rust.
- Your program cannot have memory leaks (much more C-oriented directive obviously).
- Within the mandatory part, you are allowed to use the entire libc, or the entire libstdc++, or their Rust equivalents.
- Also you can use all the functions of the math library (`-lm` man `man 3 math`)
- It is allowed to use external native libraries to open well known image formats (libpng, libjpeg ..).
- And you can use all functions of the `MinilibX` or their equivalent in another graphic library (SDL, XCB, ...): open a window, lit a pixel, put an image on a window, and manage events. The final image cannot be created by the GPU: rendering via a GPU pipeline (vertex/fragment/geometry shaders using OpenGL, Metal, Vulkan, DirectX ...) is forbidden.
- You are allowed to use other functions or other libraries to complete the bonus part (not the options) as long as their use is justified during your defence. Be smart!
- Note: GPU computing (OpenCL, CUDA, or compute shaders with OpenGL, Vulkan, Metal, ..) may be used in order to increase performances.
- You can ask your questions to other students on the forum, on slack...

Chapter V

Mandatory part

Your goal is to be able, with your program, to generate images according to the ray tracing technique.

Those computer generated images will each represent a scene, as seen from a specific angle and position of the camera / eye, defined by simple geometric objects and light sources.

This project will have a mandatory part and many additional options. The mandatory part is worth 0 points and the options will only bring you points IF the mandatory part is 100% complete. The project will only be validated if a substantial volume of options is present when you defend the project.

The elements you need to create are as follows:

- Code in C, C++ or Rust (use the latest version of the language and follow some up-to-date good practice).
- Implement the ray tracing method to create a computer generated image.
- You need at least 4 simple geometric objects as a base (not composed): plane, sphere, cylinder and cone.
- Your program must be able to apply translation and rotation transformations to objects before displaying them. For example a sphere declared at `(0, 0, 0)` must be able to successfully translate to `(42, 42, 42)`.
- Position and direction of the camera (or eye) can be changed easily.
- Manage to redraw the view or part of the view without recalculating the entire image (ex: with MinilibX, you can manage the expose event properly).
- Light management: different brightness, shadows, multi-spot, shine effect.

Chapter VI

Options

The options part will only be assessed if the mandatory part is PERFECT. Perfect means the mandatory part has been integrally done and works without malfunctioning. If you have not passed ALL the mandatory requirements, your options part will not be evaluated at all.

Now, let's get to the real deal: the options.

There are many of them, and they don't really have any limits. We can give as examples:

- Ambiance light
- Direct light
- Parallel light
- Limited objects: parallelograms, disks, half-spheres, tubes etc...
- Bump mapping and colour disruption
- External files for scene description
- Reflection
- Transparency
- Shadow modification according to transparency of the elements
- Composed elements: cubes, pyramids, tetrahedrons...
- Textures
- Negative elements
- Limit disruption / transparency / reflection, depending on texture
- More native elements: paraboloid, hyperboloid, tablecloth, toroid...

Those are the basics stuff, naturally you can be more exotic! You can think for example of multiple calculations spread over multiple computers or you can use quadrics or quartics, videos clips created from your own images, a stereoscopic version for VR

headsets etc...

Do not hesitate to be curious, explore various blogs and documentations, it actually exists many more possible options. The evaluation sheet will name various possible options, but on many items of the evaluation, you will have the possibility to give points to unlisted new options.

Note: for those of you that might want to parse some .pov or .3ds files, your ray tracer must to be able to manage correctly the simple objects of the mandatory part from equations and not from vertexes and triangles.

In addition, you have to anticipate that during the defence you will need to do some live manipulations of your scenes, using your own tools (and not existing public tools). We suggest you implement the configuration file option, or allow in-program live configuration and modifications.

In order to facilitate the evaluation of the mandatory part, we strongly advise to reproduce at least the 3 following scenes:

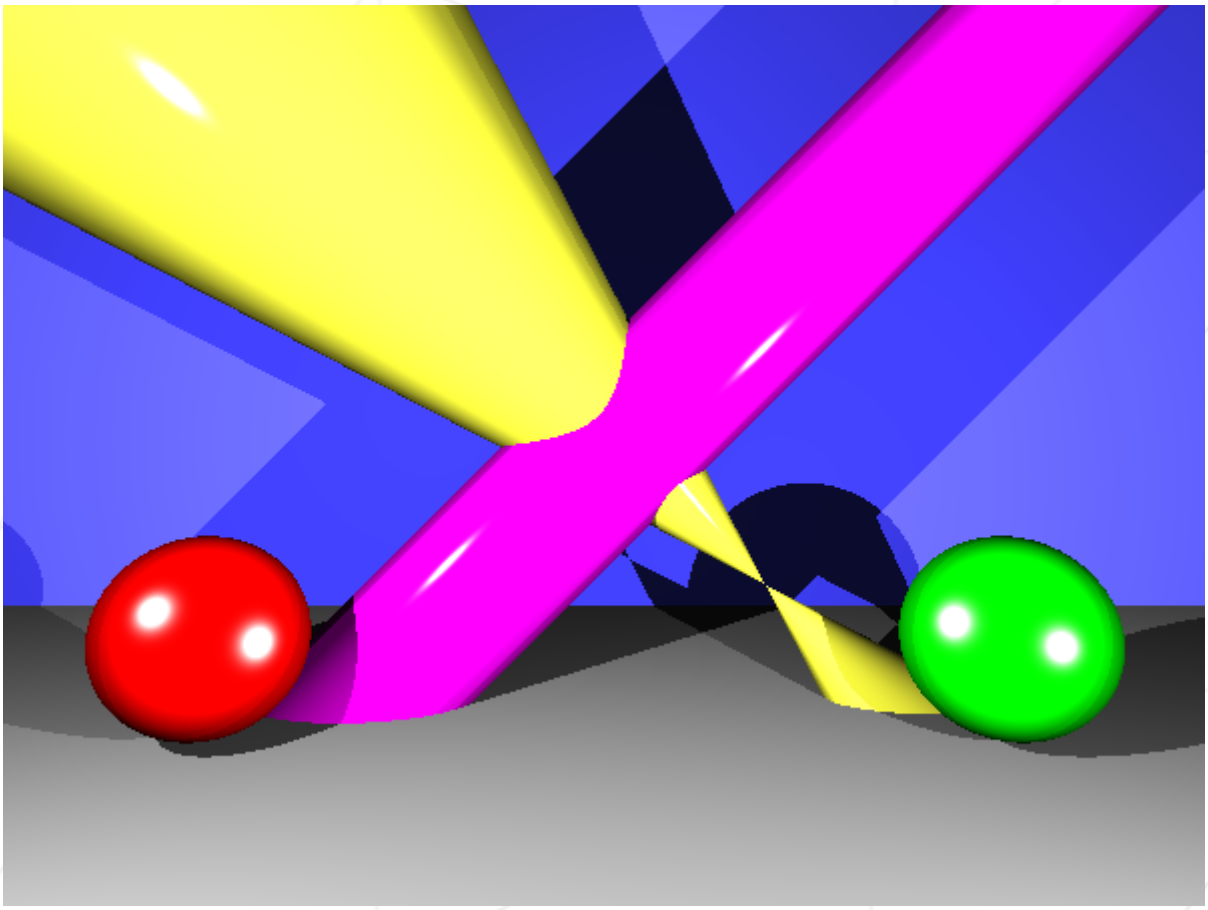

Figure VI.1: The 4 basic objects, 2 spots, shadow and shine effects

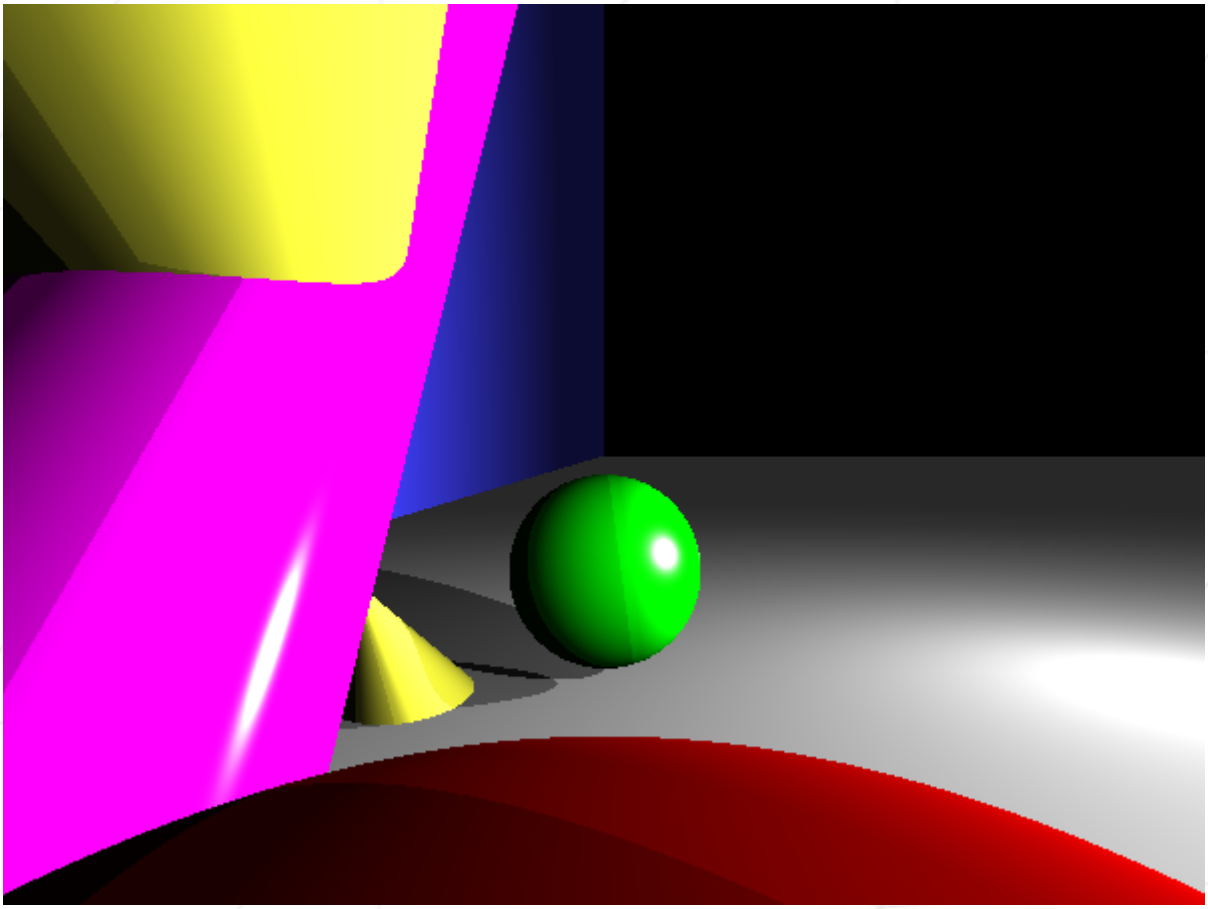

Figure VI.2: Same scene from another viewpoint

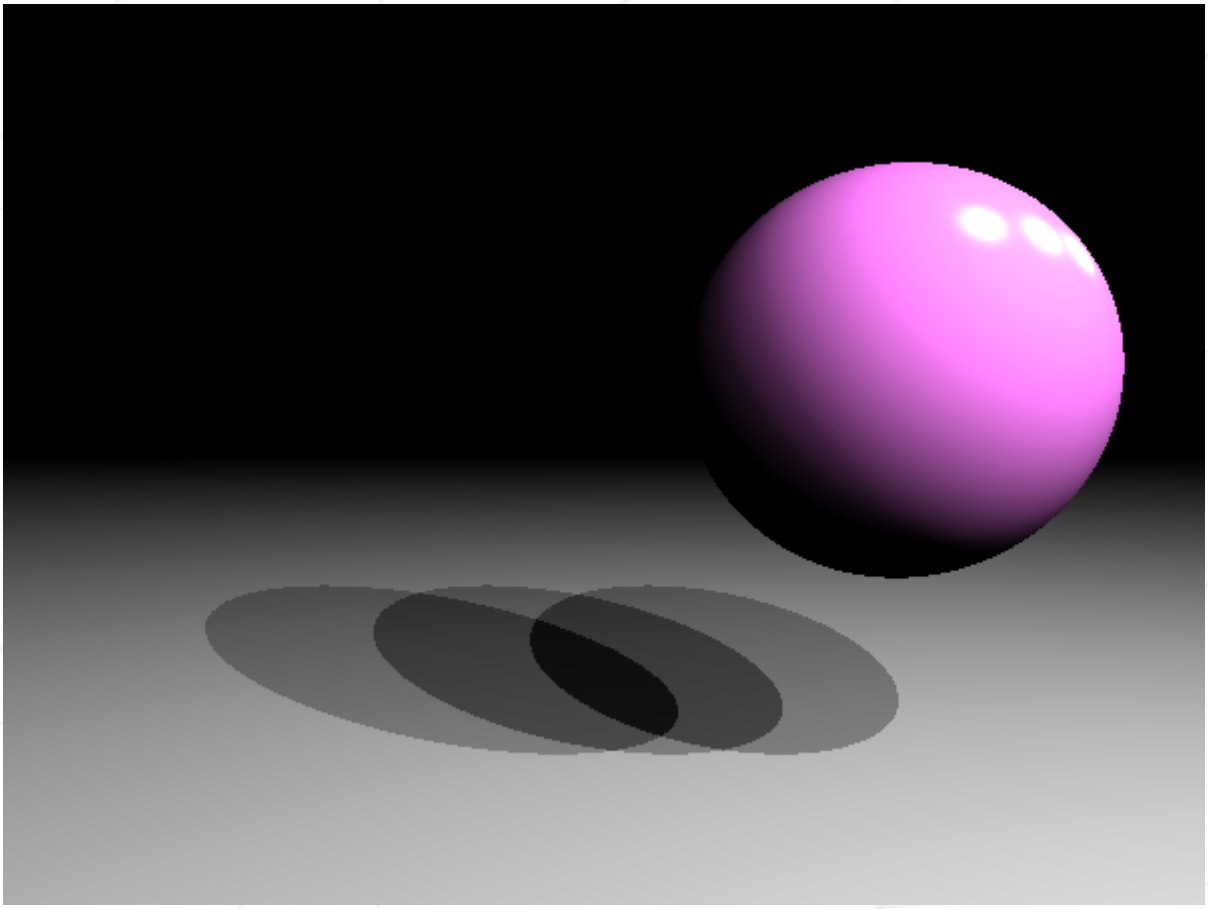

Figure VI.3: Shadow mixing

Chapter VII

Bonuses

The evaluation sheet offers bonus points for even more outstanding options, adds-on, features of your program. The final mark can go up to 125, as usual.

Chapter VIII

Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defence. Don't hesitate to double check the names of your folders and files to ensure they are correct.

You will have to demonstrate all your options during the evaluation in order to get the points. Get prepared with multiple configured scenes, ready to be calculated (already create images - jpeg, png ... - are not allowed to prove options).

The validation threshold of the project is low, the more options you create, the more XP you'll get.